

End-to-End Data Analysis — Clear Steps

1. Define the Goal

- **What do you want to find out?**
 - Example: *Which products sell best? Why do customers churn?*

2. Get the data

- Download data from **Kaggle** or other sources.

3. Load the Data

- Use **Python** with **pandas**:

4. Understand the Data (EDA: Exploratory Data Analysis)

- **Look at the shape**: rows, columns.
- **Check data types**.
- **See missing values**.
- **Basic stats**: mean, median, min, max.

5. Clean the Data

- Handle **missing values**: drop or fill with mean/median/mode.
- Fix **wrong data types**.
- Remove **duplicates**.

6. Analyze & Visualize

- Ask questions and answering it through plotting the graphs:
 - What trends do you see?
 - Any correlation?

7. Save the clean data

- Export to csv or json or excel the cleaned data

8. Make Simple Predictions (Optional)

- If you want to do Machine Learning:
 - Split into train/test.
 - Build model (e.g., Linear Regression, Random Forest).
 - Evaluate accuracy.

9. Communicate Results for End Users

This is VERY important — don't stop at coding!

- Make clear **visuals**: charts, dashboards.
- Create a **summary**:
 - “*We found that...*”
 - “*The biggest factor is...*”
- Example: “*Customers aged 18–25 spend the most. Sales increase in summer.*”